

SOC Lab 1 – Investigating Linux Authentication Logs

Objectives:

By the end of this lab, I should be able to:

- Explain what a log is.
 - Locate Linux authentication logs.
 - Generate authentication events.
 - Read log entries.
 - Identify successful logins.
 - Identify failed logins.
 - Search logs using Linux commands.
 - Count repeated events.
 - Decide whether activity is normal or suspicious.
 - Write a short investigation report.
-

Story (Real-World Scenario | 3rd person POV)

You are a Tier 1 SOC Analyst at a company.

At 11:00 AM, the Security Operations Center receives this alert:

"Multiple authentication events were detected on the Linux server overnight."

Your team lead asks:

"Determine whether these authentication events are normal user activity or something suspicious."

You have access only to the Linux server. Your job is to investigate using system logs.

Understand Concept

What is logs:

Logs is a record of events (like a receipt).

What does authentication mean?

It means to verify a user's identity before letting them access.

What kinds of authentication events can happen?

Success logins, failed login, sudo, ssh login, su

Where does Linux keep authentication events?

I am using **Kali Linux with systemd-journald (no rsyslog)**, authentication logs are stored in the **systemd journal**, not in `/var/log/auth.log`

Checking if the system-journald is running:

```
(cy@kali)-[/home/kali]
$ systemctl status systemd-journald
● systemd-journald.service - Journal Service
   Loaded: loaded (/usr/lib/systemd/system/systemd-journald.service; static)
   Active: active (running) since Thu 2026-07-02 05:56:56 EDT; 1min 29s ago
  Invocation: 11367327adb44bc9ae324c9569a25589
  TriggeredBy: ● systemd-journald-dev-log.socket
```

system is active and running and ready to be collect.

Generate authentication events (as attacker)

I open the two terminals in my linux. One to generate events, one is to investigate.

Failed local login attempts

```
(cy@kali)-[/home/kali]
$ su - nonexistentuser
su: user nonexistentuser does not exist or the user entry does not contain all the required fields
```

```
(cy@kali)-[~]
$ sudo su - root
[sudo] password for cy:
Sorry, try again.
[sudo] password for cy:
Sorry, try again.
[sudo] password for cy:
sudo: 3 incorrect password attempts
```

To test SSH, I first enabled it on my localhost and then verified that the service was running.

```
(cy@kali)-[~]
$ sudo systemctl status ssh
[sudo] password for cy:
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-07-02 07:08:52 EDT; 1h 18min ago
  Invocation: ad5baa9c575c4d5ba7ff1113c3d666ee
     Docs: man:sshd(8)
           man:sshd_config(5)
```

Failed SSH attempts

```
(cy@kali)-[~]
$ ssh baduser@localhost
baduser@localhost's password:
Permission denied, please try again.
baduser@localhost's password:
Permission denied, please try again.
baduser@localhost's password:
baduser@localhost: Permission denied (publickey,password).
```

Successful SSH login attempts

```
(cy@kali)-[~]
$ ssh cy@localhost
cy@localhost's password:
Linux kali 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25) x86_64

The programs included with the Kali GNU/Linux system are free software; the exact distribution terms for each program are described in the individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.
Last login: Thu Jul 2 06:48:46 2026 from ::1
(Message from Kali developers)

This is a minimal installation of Kali Linux, you likely want to install supplementary tools. Learn how:
=> https://www.kali.org/docs/troubleshooting/common-minimum-setup/

(Run: "touch ~/.hushlogin" to hide this message)
(cy@kali)-[~]
$
```

Sudo events (successful and failed)

```
(kali@kali)-[~]
$ su cy
Password:
(cy@kali)-[/home/kali]
$ sudo -k
(cy@kali)-[/home/kali]
$ sudo whoami
[sudo] password for cy:
root

(cy@kali)-[/home/kali]
$ sudo whoami
[sudo] password for cy:
Sorry, try again.
[sudo] password for cy:
Sorry, try again.
[sudo] password for cy:
sudo: 3 incorrect password attempts
```

I repeated these login attempts, both successful and failed, to generate logs that we can investigate later. In this way we can identify different logs and investigate the events if it is a normal activity or suspicious.

Investigation Report — Linux Auth Log Review

Date: July 03, 2026

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Summary

In this hands-on controlled lab, I intentionally generated authentication events to produce logs for analysis using Linux commands to investigate like a SOC Analyst in a pretend real-

life situations. To determine whether these authentication events are normal user activity or something suspicious.

Timeline

Example:

Timestamp	Event	Username
Jul 03 11:47:41 - 11:47:58	sudo authentication failure	-
Jul 03 11:47:59 - 11:48:11	Failed login (invalid user)	baduser
Jul 03 11:48:11 - 11:48:47	sudo authentication failure	
Jul 03 11:48:48 - 11:48:59	Failed login	cy
Jul 03 11:56:44 - 11:56:47	Successful SSH login	cy

```
Jul 03 11:47:59 kali sshd-session[7230]: Failed password for invalid user baduser from ::1 port 53888 ssh2
Jul 03 11:48:04 kali sshd-session[7230]: Failed password for invalid user baduser from ::1 port 53888 ssh2
Jul 03 11:48:11 kali sshd-session[7230]: Failed password for invalid user baduser from ::1 port 53888 ssh2
Jul 03 11:48:48 kali sshd-session[7708]: Failed password for cy from ::1 port 55182 ssh2
Jul 03 11:48:54 kali sshd-session[7708]: Failed password for cy from ::1 port 55182 ssh2
Jul 03 11:48:59 kali sshd-session[7708]: Failed password for cy from ::1 port 55182 ssh2
Jul 03 11:49:09 kali sshd-session[7930]: Accepted password for cy from ::1 port 46590 ssh2
Jul 03 11:56:11 kali sshd-session[12397]: Failed password for cy from ::1 port 56904 ssh2
Jul 03 11:56:17 kali sshd-session[12397]: Failed password for cy from ::1 port 56904 ssh2
Jul 03 11:56:21 kali sshd-session[12397]: Accepted password for cy from ::1 port 56904 ssh2
Jul 03 12:07:00 kali sshd-session[19092]: Failed password for invalid user baduser from ::1 port 56156 ssh2
Jul 03 12:07:04 kali sshd-session[19092]: Failed password for invalid user baduser from ::1 port 56156 ssh2
Jul 03 12:07:09 kali sshd-session[19092]: Failed password for invalid user baduser from ::1 port 56156 ssh2
Jul 03 12:11:10 kali sshd-session[21701]: Accepted password for cy from ::1 port 52254 ssh2
```

Findings

Total failed attempts:

```
(cy@kali)-[~]
$ sudo journalctl \
--since "2026-07-03 11:47:00" \
--until "2026-07-03 13:17:00" \
--no-pager | grep -iEc "failed"
378
```

Usernames targeted:

```
(cy@kali)-[~]
$ sudo journalctl -u ssh --no-pager | \
grep -E "Accepted password|Failed password" | \
sed -E 's/.*(Accepted password for|Failed password for) (invalid user)?([^\s]+).*/\3/' | \
sort -u
anotheruser
baduser
cy
kali
newsshuser
newuser
wronguser
```

Successful logins:

```
Jul 02 18:48:45 kali sshd-session[621663]: Accepted password for cy from ::1 port 51822 ssh2
Jul 02 18:51:33 kali sshd-session[623467]: Accepted password for kali from ::1 port 49668 ssh2
Jul 02 19:10:26 kali sshd-session[635191]: Accepted password for kali from 127.0.0.1 port 43086 ssh2
Jul 02 20:28:57 kali sshd-session[668443]: Accepted password for cy from ::1 port 45716 ssh2
Jul 02 20:32:53 kali sshd-session[672810]: Accepted password for cy from ::1 port 57168 ssh2
Jul 02 20:40:04 kali sshd-session[680800]: Accepted password for cy from ::1 port 56138 ssh2
Jul 03 11:43:43 kali sshd-session[4646]: Accepted password for cy from ::1 port 43014 ssh2
Jul 03 11:49:09 kali sshd-session[7930]: Accepted password for cy from ::1 port 46590 ssh2
Jul 03 11:56:21 kali sshd-session[12397]: Accepted password for cy from ::1 port 56904 ssh2
Jul 03 12:11:10 kali sshd-session[21701]: Accepted password for cy from ::1 port 52254 ssh2
Jul 03 13:27:21 kali sshd-session[3602]: Accepted password for cy from ::1 port 38572 ssh2
Jul 03 13:32:01 kali sshd-session[6587]: Accepted password for cy from ::1 port 57628 ssh2
```

All source IP(s): 127.0.0.1 (this is only a test)

Assessment

Since I generated different login attempt, the logs results show rapid authentication login attempts within a short period. These include multiple failed or success login attempts where both valid and non-existent usernames is from a single IP address (127.0.0.1) that targeting accounts. This pattern resembles a password guessing or brute-force attack. Even though the activity is from localhost as part of a controlled lab, the severity is medium because this can be a similar behavior of malicious activity. Therefore, to determine whether the successful logins were authorized or not, this log analysis should be flagged as suspicious and needs further investigation.

Evidence

Let's check in the journalctl if logs was generate to verify that the events were successfully recorded.

```
Jul 03 13:27:03 kali sshd-session[3406]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=::1
Jul 03 13:27:03 kali sshd-session[3406]: Failed password for invalid user baduser from ::1 port 54554 ssh2
Jul 03 13:27:08 kali sshd-session[3406]: Failed password for invalid user baduser from ::1 port 54554 ssh2
Jul 03 13:27:12 kali sshd-session[3406]: Failed password for invalid user baduser from ::1 port 54554 ssh2
Jul 03 13:27:12 kali sshd-session[3406]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=::1
Jul 03 13:27:21 kali sshd-session[3602]: Accepted password for cy from ::1 port 38572 ssh2
Jul 03 13:27:38 kali sudo[3839]: pam_unix(sudo:auth): authentication failure; logname=cy uid=1001 euid=0 tty=/dev/pts/1 ruser=cy rhost= user=cy
Jul 03 13:27:47 kali sudo[3839]: cy : 3 incorrect password attempts ; TTY=pts/1 ; PWD=/home/cy ; USER=root ; COMMAND=/usr/bin/whoami
Jul 03 13:32:01 kali sshd-session[6587]: Accepted password for cy from ::1 port 57628 ssh2
```

Confirmed it collects and records the event and shows authentication event

Success:

```
Jul 03 11:43:43 kali sshd-session[4646]: Accepted password for cy from ::1 port 43014 ssh2
Jul 03 11:49:09 kali sshd-session[7930]: Accepted password for cy from ::1 port 46590 ssh2
Jul 03 11:56:21 kali sshd-session[12397]: Accepted password for cy from ::1 port 56904 ssh2
Jul 03 12:11:10 kali sshd-session[21701]: Accepted password for cy from ::1 port 52254 ssh2
Jul 03 13:27:21 kali sshd-session[3602]: Accepted password for cy from ::1 port 38572 ssh2
Jul 03 13:32:01 kali sshd-session[6587]: Accepted password for cy from ::1 port 57628 ssh2

(cy@kali)-[~]
$ sudo journalctl --no-pager | grep -iE "Accepted" | wc -l
12
```


Failures (failed password/incorrect password):

```
ruser= rhost=:1
Jul 03 13:27:03 kali sshd-session[3406]: Failed password for invalid user baduser from ::1 port 54554 ssh2
Jul 03 13:27:08 kali sshd-session[3406]: Failed password for invalid user baduser from ::1 port 54554 ssh2
Jul 03 13:27:12 kali sshd-session[3406]: Failed password for invalid user baduser from ::1 port 54554 ssh2
Jul 03 13:27:12 kali sshd-session[3406]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= r
host=:1
Jul 03 13:27:38 kali unix_chkpwd[3858]: password check failed for user (cy)
Jul 03 13:27:38 kali sudo[3839]: pam_unix(sudo:auth): authentication failure; logname=cy uid=1001 euid=0 tty=/dev/p
ts/1 ruser=cy rhost= user=cy
Jul 03 13:27:40 kali unix_chkpwd[3888]: password check failed for user (cy)
Jul 03 13:27:45 kali unix_chkpwd[3934]: password check failed for user (cy)
Jul 03 13:27:47 kali sudo[3839]: cy : 3 incorrect password attempts ; TTY=pts/1 ; PWD=/home/cy ; USER=root ;
COMMAND=/usr/bin/whoami
Jul 03 13:27:51 kali (script.sh)[4001]: mytimer.service: Failed at step EXEC spawning /full/path/to/my/script.sh: N
o such file or directory
Jul 03 13:27:51 kali systemd[1]: mytimer.service: Failed with result 'exit-code'.
```

Su/sudo outcomes if authentication failure or session opened

```
d=0)
Jul 03 13:26:48 kali su[3267]: pam_unix(su-l:auth): authentication failure; logname=cy uid=1001 euid=0 tty=/dev/pts
/0 ruser=cy rhost= user=root
Jul 03 13:27:01 kali sshd-session[3406]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh
ruser= rhost=:1
Jul 03 13:27:12 kali sshd-session[3406]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= r
host=:1
Jul 03 13:27:21 kali sshd-session[3602]: pam_unix(sshd:session): session opened for user cy(uid=1001) by cy(uid=0)
Jul 03 13:27:38 kali sudo[3839]: pam_unix(sudo:auth): authentication failure; logname=cy uid=1001 euid=0 tty=/dev/p
ts/1 ruser=cy rhost= user=cy
Jul 03 13:27:55 kali sudo[3839]: pam_unix(sudo:auth): authentication failure; logname=cy uid=1001 euid=0 tty=/dev/p
ts/1 ruser=cy rhost= user=cy
```

Failed login attempts by username in ssh

```
sort | uniq -c | sort -rn
 27 baduser
   8 cy
   6 wronguser
   6 newuser
   3 newsshuser
   2 kali
   1 anotheruser
```

Failed login attempts by source IP

```
(cy@kali)-[~]
$ sudo journalctl -u ssh --no-pager | \
grep "Failed password" | \
grep -oE 'from ([0-9]{1,3}\.){3}[0-9]{1,3}' | \
awk '{print $2}' | \
sort | uniq -c | sort -rn
 5 127.0.0.1
```

Successful logins by username

```
(cy@kali)-[~]
$ sudo journalctl -u ssh --no-pager | \
grep "Accepted password" | \
sed -E 's/.*Accepted password for ([^ ]+) from.*/\1/' | \
sort | uniq -c | sort -rn
 10 cy
   2 kali
```

Recommendation

No action needed, this only a self-testing. However in a real-life scenario, cybersecurity team should use a firewall or IDS/IPS to block repeated failed login attempts from suspicious sources. Enforcing password policy such as making a stronger password, add multi-factor authentication for another verification, and limit logging attempts by locking accounts after

after a set number of failed. Set up SIEM rules for monitoring and alerts security analyst. Lastly, conduct regular log audits to review, identify trends or anomalies, and potential insider threats.